

22. Pownal Bay

Shoreline length: 33 km



Shore Types

Pownal Bay is located on the southeast end of PEI. Shore type characterized by a majority of wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
13	4	1	8	74	0	0	0



Bluff off Earnscliffe road south of Pownal Bay



Cherrycliff Park and Earnscliffe beach



Coastal wetlands along Cherry Valley

Governance, Infrastructure, and Economy

This area is unincorporated. Infrastructure includes Pownal Road (Route 26, at Mount Albion Creek).

There are no key coastal economic drivers within the coastal section of this region. Local and legacy land use constraints include several small

undeveloped lots, accessed by unpaved or undeveloped coastal roads.

Natural Assets and Heritage

There are two main recreation assets: Cherrycliff Park (playground and beach access) and the Pownal Sports Centre (double ice pad, a 600-seat stadium, a 150-seat arena, and community rooms).

The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is low.

Coastal Flood and Erosion Risk

Infrastructure within in this area are vulnerable to increasing flood hazards. Erosion hazard is highest along the west facing cliffs with limited development. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	0	0	1 [<1/km]	4 [<1/km]	0	1 [<1/km]	4 [<1/km]
Roads	458 m [14 m/km]	590 m [18 m/km]	749 m [22 m/km]	1526 m [46 m/km]	13 m [0 m/km]	56 m [2 m/km]	201 m [6 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
22.00 Pownal Bay	NAI	MR	MR
Rationale	Given the initially low risk levels the proposed strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets while preserving natural shorelines.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.